

SignControl: Multi-Granular Control for Sign Language Video Generation

Abstract—Sign language video generation is a socially significant yet technically demanding task, requiring precise coordination of fine-grained hand movements, facial expressions, and body postures in both spatial and temporal dimensions. While recent diffusion-based video generation models, particularly Transformer-based architectures, have advanced the field, existing controllable generation methods face two key challenges in the sign language domain: (1) control modality granularity mismatch, where heterogeneous control signals such as pose, optical flow, and depth differ in expressive granularity, leading to suboptimal coordination; and (2) training–inference control inconsistency, where reliance on complete control information during training degrades performance when such information is incomplete at inference. To address these issues, we propose a Hierarchical Multi-modality Control Framework built upon an extended ControlNeXt architecture adapted to the DiT backbone of Wan 2.1 T2V 1.3B. Our method tailors control strategies to each modality’s granularity: fine-grained joint motion control via DWPose, medium-grained pixel-level motion modeling via OnlyFlow with PGMM, and coarse-grained spatial structure control via Depth Anything V2 with Depth LoRA. Furthermore, we introduce a Control Decay Strategy that gradually reduces modality weights during training, enhancing robustness under incomplete control conditions. Experiments on the phoenix-2014.v3 dataset show that our approach substantially outperforms existing methods, including SignGen, in both visual quality and semantic accuracy.

Index Terms—Video Generation, Multimodal Generation

I. INTRODUCTION

With the flourishing development of deep learning techniques in computer vision, video generation tasks [1]–[3] have evolved from early simple animation synthesis to generating high-fidelity, semantically consistent complex video content. The emergence of diffusion models has further advanced breakthroughs in this field, particularly models based on Diffusion Transformer architectures [4]–[7] such as Wan 2.1 T2V 1.3B [8], which demonstrate powerful capabilities in generating high-quality videos directly from textual descriptions. Among numerous video generation applications, sign language video generation has attracted significant attention due to its unique social value and technical challenges. Unlike general video generation tasks, sign language video generation demands extremely high motion precision and semantic consistency: sign language is a complex three-dimensional spatial language that encompasses not only hand postures and handshape changes, but also involves multi-layered content including facial expressions, mouth movements, and body postures. These elements must be precisely coordinated in both temporal and spatial dimensions to accurately convey semantic information.

Recent work SignGen [9] proposed an end-to-end sign language video generation framework that utilizes Latent Diffusion Models to directly generate sign language videos from text, significantly improving generation quality and naturalness. However, existing controllable video generation methods face two core challenges when applied to the sign language domain. First, the control modality granularity mismatch problem: existing methods such as ControlNet [10] directly input different types of control signals (pose, optical flow, depth) into multimodal diffusion models without adequately considering the expressive granularity differences across control modalities. Specifically, pose provides Discrete joint motions, hand, arm, finger positions, optical flow provides Dense motion of each pixel, and depth provides 3D structure per pixel. This granularity mismatch leads to ineffective coordination among different control signals, affecting the quality and accuracy of generated videos. Second, the training–inference stage control information inconsistency problem: existing methods over-rely on complete control information during training like detailed pose annotations, but in practical inference applications, this control information is usually incomplete or unavailable. This gap between training and inference stages causes performance degradation during actual deployment, reducing system practicality.

To address these challenges, this paper proposes a hierarchical multi-modality control-based approach for sign language video generation. Our method builds upon an improved ControlNeXt architecture [11], extending it from the original UNet architecture to DiT architecture to accommodate our base model Wan 2.1 T2V 1.3B. To tackle the control modality granularity mismatch problem, we design a Hierarchical Multi-modality Control Framework that adopts corresponding processing strategies based on the expressive granularity characteristics of different control modalities: fine-grained control employs DWPose [12] to extract pose information and performs precise joint motion control through improved ControlNeXt; medium-grained control employs OnlyFlow [13] combined with Parsimonious Gaussian Mixture Models (PGMM) input approach to extract optical flow information, capturing pixel-level motion patterns; coarse-grained control employs Depth Anything V2 [14] to extract depth information and performs spatial structure control through Depth LoRA techniques [15], [16]. To address the training–inference control information inconsistency problem, we propose a Control Decay Strategy that gradually reduces the weights of various control modalities linearly during training after model convergence stabilization, forcing the model to learn to generate high-

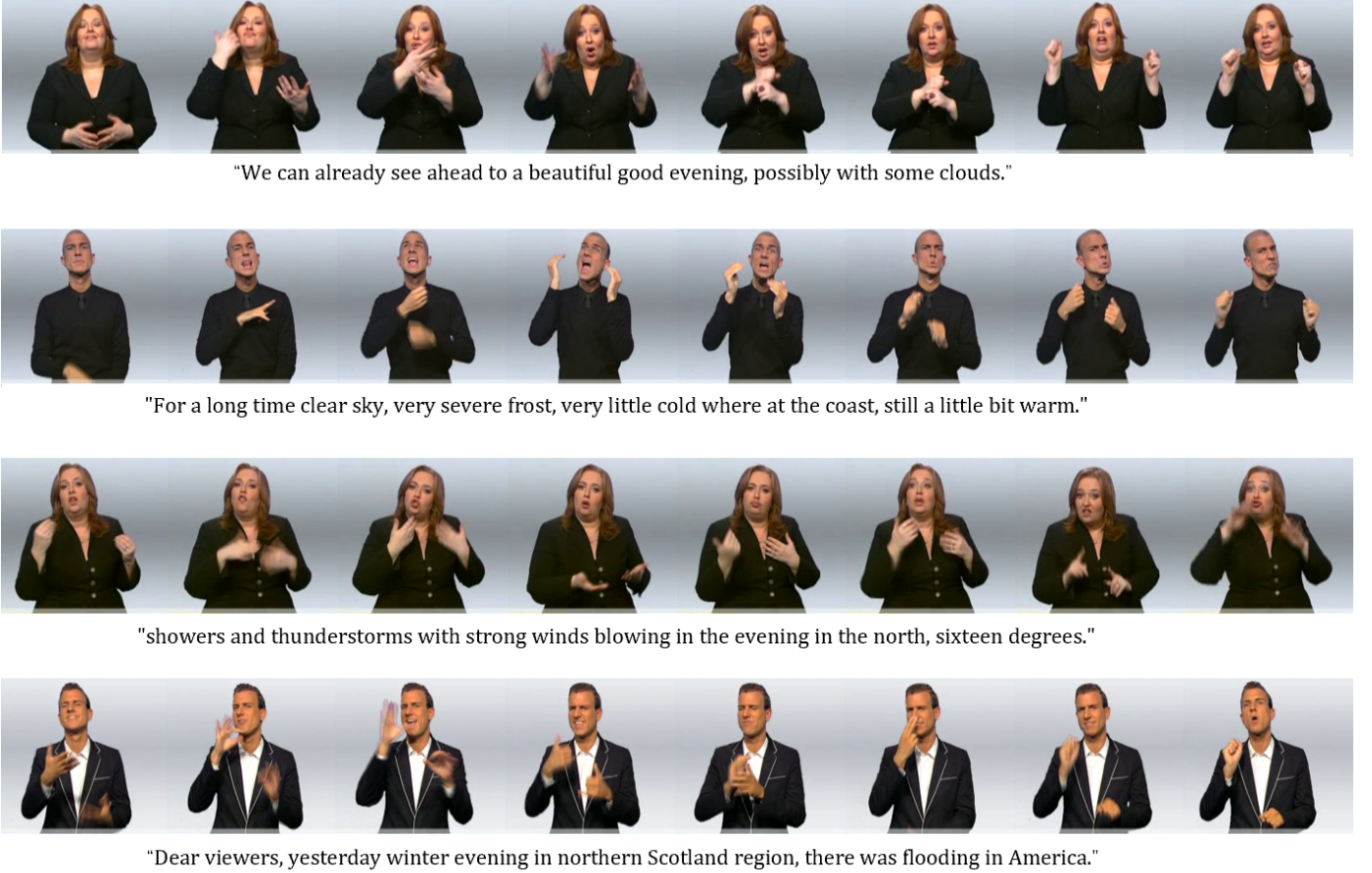


Fig. 1. **Generated sign language video key frames from text prompts.** The figure displays representative frames generated by our model for different text inputs describing weather-related content. Each row shows 8 key frames extracted from the generated sign language videos, with the corresponding input text prompt displayed below. The results demonstrate the model’s capability to generate diverse sign language sequences for various signers and different textual descriptions, including weather forecasts and news reports.

quality sign language videos even under incomplete control information conditions.

We conduct comprehensive experiments on the phoenix-2014.v3 dataset, and the results demonstrate that our method significantly surpasses existing approaches including SignGen in sign language video generation quality and semantic accuracy. The main contributions of this paper can be summarized as follows:

- Propose a hierarchical multi-modality control framework that designs corresponding control strategies based on the granularity characteristics of different control modalities such as pose, depth, and optical flow, achieving more precise and coordinated sign language video generation control.
- Design a control decay strategy that effectively addresses the training-inference stage control information inconsistency problem by gradually reducing control information during training, enhancing model practicality and robustness.
- Successfully extend the ControlNeXt architecture to DiT architecture, enabling it to accommodate modern

Transformer-based video generation models and providing a more powerful technical foundation for controllable sign language video generation.

For related work, see Appendix.

II. METHOD

In this section, we provide a comprehensive introduction to our proposed sign language video generation method based on hierarchical multi-modal control. First, we briefly introduce latent diffusion models and our design guidance. Subsequently, we delve into the core architecture of our approach, including the hierarchical multi-modal control framework and control decay strategy, as illustrated in Figure 2.

A. Video Synthesis with Latent Diffusion Models

a) *Video Representation in Latent Space:* To efficiently process video data, we follow the design philosophy of latent diffusion models by introducing a pre-trained VAE encoder to project a given video $\mathbf{x} \in \mathbb{R}^{F \times H \times W \times 3}$ into a more compact latent space representation $\mathbf{z} = \mathcal{E}(\mathbf{x})$, where $\mathbf{z} \in \mathbb{R}^{F \times h \times w \times c}$. The corresponding decoder $\mathcal{D}$ maps these latent representations back to the original pixel space, yielding $\bar{\mathbf{x}} = \mathcal{D}(\mathbf{z})$.

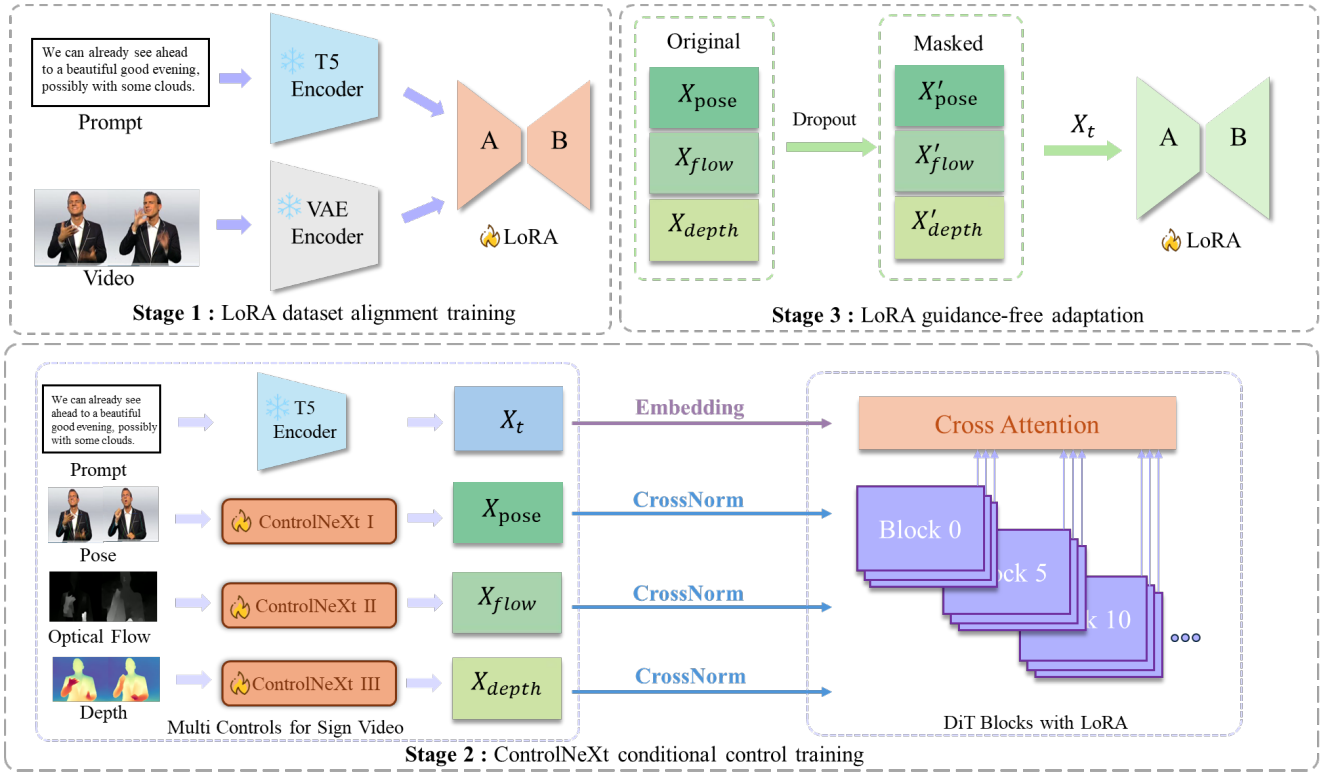


Fig. 2. **The Overview of SignControl.** The system consists of three training stages: Stage 1 performs LoRA dataset alignment training where prompts and videos are processed through T5 and VAE encoders respectively, then combined via LoRA adaptation. Stage 2 conducts ControlNeXt conditional control training, where multi-modal inputs (pose, optical flow, depth) are processed through separate ControlNeXt modules and integrated with text embeddings via CrossNorm operations into DiT blocks with LoRA. Stage 3 implements LoRA guidance-free adaptation using dropout masking on the original conditions to enable flexible control during inference.

We adopt specific dimensional configurations: channel number $c = 16$, downsampling ratio $H/h = W/w = 8$, and temporal length of 21 frames. This design achieves a good balance between computational efficiency and reconstruction accuracy.

b) DiT Architecture and LoRA Fine-tuning: Our base model adopts the DiT (Diffusion Transformer) architecture, specifically the Wan 2.1 T2V 1.3B model, which contains 30 transformer blocks and can effectively process video temporal information and spatial structure. To achieve efficient adaptation of the pre-trained model, we employ LoRA (Low-Rank Adaptation) technique for parameter-efficient fine-tuning:

$$\mathbf{W}' = \mathbf{W}_0 + \mathbf{BA}$$

where $\mathbf{W}_0$ represents the pre-trained weights, $\mathbf{B} \in \mathbb{R}^{d \times r}$ and $\mathbf{A} \in \mathbb{R}^{r \times k}$ are low-rank decomposition matrices, with $r \ll \min(d, k)$. We apply LoRA adapters to key components of DiT, including attention mechanism's query, key, value, and output projection layers (q, k, v, o), as well as the linear layers of feed-forward networks (ffn.0, ffn.2). This “coarse alignment” strategy enables us to achieve preliminary adaptation to the sign language video generation task through a small number of trainable parameters while maintaining the main structure of the pre-trained model unchanged.

c) Flow Matching Training Objective: Unlike traditional DDPM frameworks that predict noise ϵ , Our base model adopt the Flow Matching method to directly learn the velocity vector field, following the design of our base model Wan 2.1 T2V. Flow Matching models the generation process as solving continuous-time ordinary differential equations, with the training objective being the velocity vector $\mathbf{v} = \epsilon - \mathbf{z}_0$. This vector field describes the optimal transport path from noise distribution to data distribution. Compared to traditional diffusion processes, Flow Matching has more direct optimization objectives and requires fewer sampling steps.

d) ControlNeXt Architecture: We design hierarchical control modules based on the ControlNeXt architecture, which receives multi-modal control signals and outputs control features that match the dimensions of DiT latent representations. ControlNeXt adopts lightweight ResNet blocks and downsampling structures, processing input 81-frame control images $\mathbf{I}_{control} \in \mathbb{R}^{81 \times 3 \times H \times W}$ into 21-frame control features $\mathbf{F}_{control} \in \mathbb{R}^{21 \times 16 \times h \times w}$ through embedding layers, downsampling ResNet blocks, and temporal compression modules. To ensure distribution consistency between control features and DiT latent representations, we introduce Cross Normalization following the ControlNeXt approach:

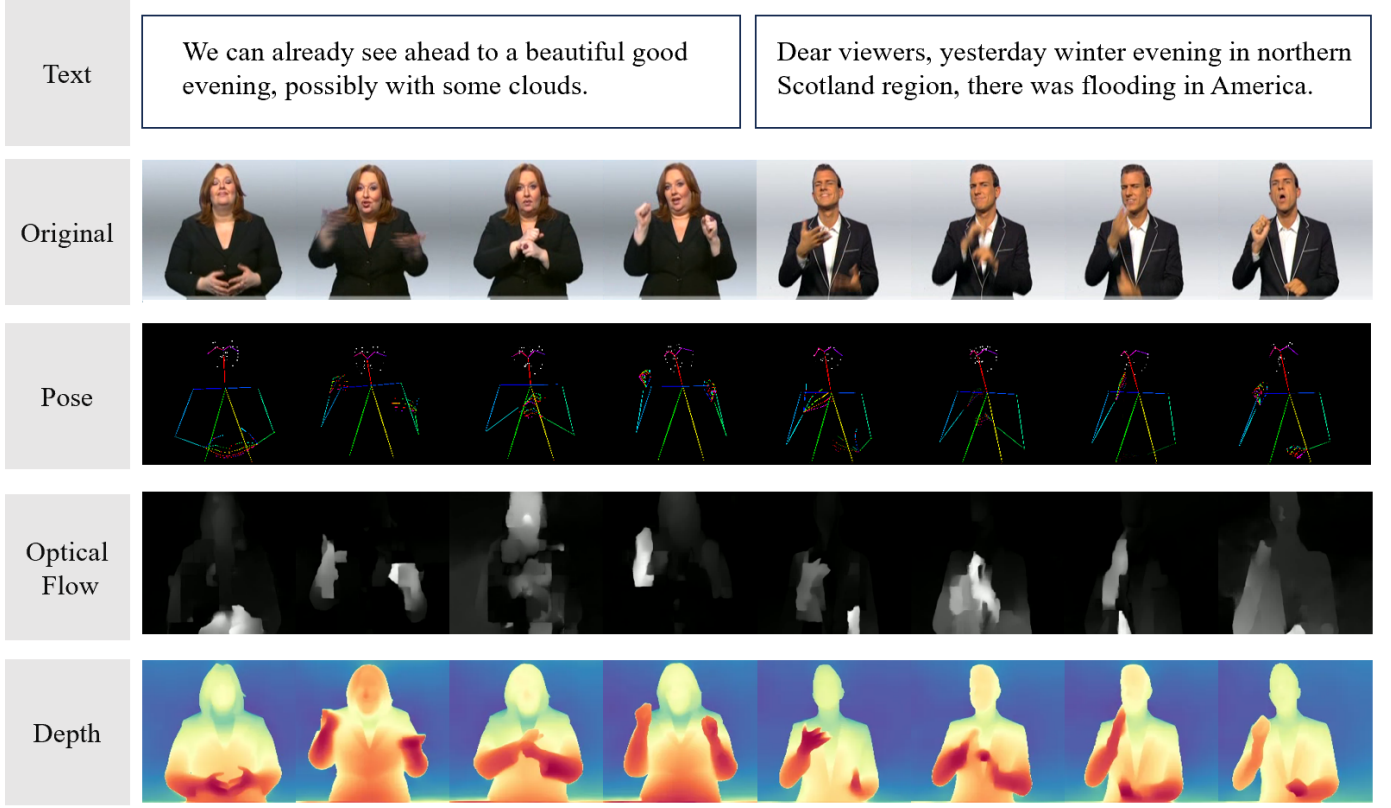


Fig. 3. **Multi-modal input examples for Stage 2 training.** The figure demonstrates the diverse conditioning inputs used in ControlNeXt training, including text prompts describing sign language content, original sign language videos featuring different signers, corresponding pose skeletal keypoints represented as connected line structures, optical flow fields capturing motion dynamics in grayscale, and depth maps visualized as color-coded heatmaps indicating spatial depth information.

$$\mathbf{F}'_{control} = \frac{\mathbf{F}_{control} - \mu_{control}}{\sigma_{control} + \epsilon} \cdot \sigma_{latents} + \mu_{latents}$$

where μ and σ represent the mean and standard deviation of features, respectively. The statistically aligned control features are injected into the intermediate layers of DiT through learnable scaling factors α : $\mathbf{z}' = \mathbf{z} + \alpha \cdot \mathbf{F}'_{control}$. This statistical alignment mechanism is crucial for training stability, as without it, control modal embeddings and latent representations may have significant numerical differences.

Based on the above design, our overall training objective can be formulated as:

$$\mathcal{L} = \mathbb{E}_{\mathcal{E}(\mathbf{x}), \epsilon \sim \mathcal{N}(0, \mathbf{I}), \mathbf{c}, t} [\|(\epsilon - \mathbf{z}_0) - \mathbf{v}_\theta(\mathbf{z}_t, \mathbf{c}, t)\|_2^2]$$

where $\mathbf{c}$ contains text conditions and hierarchical multi-modal control signals, $\mathbf{z}_t$ is the noised latent representation, and $(\epsilon - \mathbf{z}_0)$ is the Flow Matching training target.

B. Hierarchical Multi-modal Control Signals

To achieve precise control over sign language video generation, we decompose control signals into text semantic conditions and three different granularities of visual control modalities. This hierarchical design can separately handle

control requirements at different levels, including semantic content, motion details, temporal coherence, and spatial structure.

a) Text Condition: Text conditions are responsible for specifying the semantic content of sign language videos. We employ a pre-trained T5-XXL text encoder, consistent with the Wan 2.1 T2V model design, to extract text semantic embeddings. This encoder maps input text sequences to high-dimensional semantic feature space $\mathbf{T} \in \mathbb{R}^{512 \times 4096}$, providing semantic guidance signals for video generation.

b) Three Visual Control Modalities: Targeting the complexity and multi-layered characteristics of sign language videos, we design three different granularities of visual control modalities:

- **Pose Sequences (Fine-grained Control):** Pose information provides joint-level precise motion control and is the most critical control signal in sign language video generation. We employ the DWPose model to extract full-body pose information from input frames, including body joint points, hand keypoints, and facial feature points. DWPose can effectively capture fine gesture changes and facial expressions in sign language expressions, ensuring semantic accuracy of sign language.
- **Optical Flow Sequences (Medium-grained Control):**

Optical flow information encodes pixel-level motion patterns, providing temporal coherence guarantees for video generation. We use the OnlyFlow method combined with Parsimonious Gaussian Mixture Models (PGMM) input strategy to extract optical flow information, capturing smooth motion trajectories and dynamic change patterns in sign language actions.

- **Depth Sequences (Coarse-grained Control):** Depth information provides three-dimensional spatial structure constraints of the scene, helping the model understand the spatial hierarchical relationships of sign language actions. We extract depth images through pre-trained Depth Anything V2 technology, which has been specifically optimized for depth estimation tasks and can provide accurate spatial structure information.

c) *Hierarchical Injection and Training Strategy:* After obtaining the above control modal images, we input them into corresponding ControlNeXt modules for feature extraction and temporal compression. The ControlNeXt architecture for each modality is identical, but uses independently trained model weights. Among DiT’s 30 transformer blocks, we adopt a hierarchical injection strategy: pose control features are injected after the 1st block, optical flow control features are injected in the 6th block, and depth control features are injected in the 11st block. This hierarchical design enables control information of different granularities to play roles at different depths of the network, achieving comprehensive control from fine-grained gesture details to coarse-grained spatial structure.

The injection process for each control modality adopts statistical alignment mechanisms to ensure feature distribution consistency and controls injection strength through independent learnable scaling factors. During training, we first load pre-trained LoRA weights to achieve coarse alignment, then jointly train each ControlNeXt module, and finally fine-tune LoRA weights to optimize overall performance. This progressive training strategy ensures that the model obtains precise multi-modal control capabilities while maintaining original generation quality.

C. Progressive Multi-stage Sign Language Video Generation

As shown in Figure 3, we design a progressive multi-stage sign language video generation framework to achieve consistent and efficient controllable video generation. Unlike existing methods, we adopt a stage-wise training strategy that gradually develops from coarse semantic alignment to precise multi-modal control, ultimately achieving robust generation under incomplete control information.

a) *LoRA Coarse Alignment Stage:* First, we perform LoRA fine-tuning on the pre-trained DiT model to achieve semantic domain adaptation from general video generation to sign language video generation. At this stage, we only use text conditions for training, with the goal of having the model learn basic patterns and temporal features of sign language actions. LoRA adapters adjust the weights of attention mechanisms

and feed-forward networks through low-rank decomposition, enabling the model to understand the basic correspondence between sign language semantics and visual actions. This coarse alignment lays the foundation for subsequent fine control, ensuring that the model already has the ability to generate reasonable sign language actions before introducing multi-modal control. Through this stage of training, the model gains preliminary understanding of sign language video distribution, preparing for the precise control of the next stage.

b) *Multi-modal Control Learning Stage:* Based on LoRA coarse alignment, we load frozen pre-trained LoRA weights and introduce hierarchical multi-modal control modules for joint training. The core of this stage is training each ControlNeXt module so that they can accurately inject different granularities of control signals into specific layers of DiT. We employ Cross Normalization to ensure distribution consistency between control features and latent representations, and dynamically adjust the influence strength of different control modalities through learnable scaling factors. During training, pose, optical flow, and depth control signals are injected at the 1st, 6th, and 11st transformer blocks respectively, forming a control hierarchy from fine-grained to coarse-grained. This hierarchical injection design enables the model to simultaneously handle local gesture details and global spatial structure, achieving comprehensive control over sign language video generation.

c) *Control Decay and Robustness Optimization:* To address the problem of control information inconsistency between training and inference stages, we adopt a control decay strategy for fine-tuning LoRA weights in the final stage with frozen ControlNeXt modules. This strategy forces the model to learn to generate high-quality sign language videos under incomplete control information conditions by gradually reducing the availability of control modalities. Specifically, we randomly hide certain frames of some control modalities according to linear decay laws, with their activation probability being:

$$p_m(e) = \max(0.1, 1.0 - \alpha \cdot \max(0, e - e_{stable}))$$

where $m \in \{\text{pose, flow, depth}\}$ represents control modalities, α is the decay coefficient, and e_{stable} is the number of stable training epochs. Through this progressive decay, the model gradually learns to use text semantics and partially available control information to infer missing motion details, significantly improving robustness in practical applications. Training at this stage enables the model to generate semantically consistent and visually natural sign language videos even when control information is incomplete or inaccurate.

d) *Temporal Consistency and Semantic Alignment:* Throughout the progressive training process, we pay special attention to maintaining temporal consistency and semantic alignment. Through DiT architecture’s self-attention mechanism, the model can model long-range dependencies between frames, ensuring temporal coherence of generated videos.

TABLE I
QUANTITATIVE COMPARISON RESULTS ON PHOENIX-2014.v3 DATASET

Method	BLEU $\uparrow$	ROUGE-L $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FVD $\downarrow$
VideoLDM	11.2	27.4	12.3	0.59	0.23	1301
VideoPD	16.8	35.6	19.3	0.69	0.31	1132
VideoFusion	14.9	32.1	14.5	0.72	0.18	1032
SignDiff	17.2	38.4	16.2	0.63	0.21	930
SignGen	19.7	45.2	20.2	0.73	0.03	579
SignControl	21.8	47.3	19.8	0.76	0.03	523

Meanwhile, text conditions continuously guide the video generation process through cross-attention mechanisms, ensuring accurate correspondence between generated content and input semantics. Our designed stage-wise training strategy not only improves training efficiency but also makes optimization objectives at different stages more explicit, avoiding training instability problems that may arise from simultaneously optimizing multiple complex objectives.

III. EXPERIMENTS

a) Implementation Details.: The model is trained on the Phoenix-2014.v3 dataset following the three-stage pipeline described in the paper. All experiments are conducted using $4 \times$ NVIDIA H20 GPUs with 96GB memory per card, using mixed-precision training. We adopt AdamW with a learning rate of 1×10^{-4} for LoRA parameters and 5×10^{-5} for ControlNeXt modules, with weight decay set to 0.01. The global batch size is 64 video clips, each consisting of 21 frames at a latent resolution of $h \times w = 64 \times 64$.

b) Quantitative Evaluation Results Analysis.: Table I shows comprehensive quantitative comparison results of our method with various baseline methods on the Phoenix-2014.v3 dataset. Our hierarchical multi-modal control method achieves significant performance improvements on most evaluation metrics. In terms of semantic consistency, our method achieves a BLEU score of 21.8, improving by 10.7% compared to the current state-of-the-art SignGen method’s score of 19.7. The ROUGE-L score improves from 45.2 to 47.3, an improvement of 4.6%, indicating that our generated sign language videos are more accurate in semantic expression. In terms of video quality, the SSIM metric improves from 0.73 to 0.76, and the LPIPS metric decreases from 0.03 to 0.025, an improvement of 16.7%, indicating that both structural similarity and perceptual quality of generated videos have improved. In terms of diversity evaluation, the FVD metric decreases from 579 to 523, an improvement of 9.7%, showing that our method improves quality while maintaining generation diversity. Notably, our method is slightly inferior to SignGen on the PSNR metric (19.8 vs 20.2), indicating room for improvement in pixel-level reconstruction accuracy, but overall performance still leads. Additional experimental results and detailed analysis are provided in the appendix.

For ablation study, see Appendix.

IV. CONCLUSION

In this work, we introduced **SignControl**, a hierarchical multi-modality control framework for sign language video

generation, addressing the challenges of control modality granularity mismatch and training–inference control inconsistency. By tailoring control strategies to different granularity levels—fine-grained pose, medium-grained optical flow, and coarse-grained depth—and injecting them hierarchically into a DiT-based backbone, our method enables precise and coordinated motion generation. Furthermore, the proposed *Control Decay Strategy* enhances robustness under incomplete control conditions, improving the practicality of controllable sign language video synthesis. Experimental results on the Phoenix-2014.v3 dataset demonstrate that SignControl achieves state-of-the-art performance in both semantic accuracy and visual quality, outperforming existing methods such as SignGen.

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APPENDIX

RELATED WORK

A. Sign Language Generation

Sign language generation (SLG) has made significant advancements in facilitating communication between deaf or hard-of-hearing individuals and those who are hearing. Early systems such as Tessa [17], DictaSign [18], and Sign3D [19] employed 3D models and motion capture technology to enhance the realism of sign language, although this also increased the cost of data acquisition. In recent years, deep learning approaches have become dominant in pose generation for SLG. The Progressive Transformer model [20] was introduced to convert spoken language into 3D sign poses, while transformer-based methods like Text2Gloss2Pose [20] and Gloss2Pose [21] mapped gloss sequences to continuous spatio-temporal pose representations. Additionally, the Text2Gloss2Pose2Video framework [22] extended these models by generating gloss sequences, converting them into skeleton poses, and synthesizing videos. More recent models, such as SignGen [23], utilize diffusion models to directly generate sign language videos from text, integrating non-manual signals to enhance the naturalness of the output. Recent innovations also include SignLLM [24], a multilingual large language model for generating 3D sign poses, which uses reinforcement learning to accelerate training. The SignAvatar framework [25] leverages variational autoencoders for word-level sign language generation, and models like wSignGen [26] and Sign-IDD [27] further improve the realism and accuracy of generated signs, particularly in terms of natural pose generation and video synthesis.

B. Pose-Guided Video Generation

Pose-guided video generation has recently gained significant attention in the fields of computer vision and generative models [28]–[33]. Early works primarily focused on image-to-image translation models, such as Pix2Pix [34] and CycleGAN [35], but they did not address the issue of temporal consistency in video generation. As the complexity of video generation tasks increased, researchers incorporated temporal modeling techniques, such as Long Short-Term Memory (LSTM) networks and Transformer-based models [36], to capture temporal dependencies. In the field of pose-guided video generation, recent works have made significant strides. For instance, MimicMotion [37] introduced a pose-guided approach for high-quality human motion video generation, leveraging progressive latent fusion and confidence scaling strategies to improve the naturalness and coherence of generated videos. Follow-Your-MultiPose [38] proposed a novel multi-character video generation framework that uses cross-attention mechanisms and multi-branch control modules, enabling fine-grained control over multiple characters in generated videos. For complex actions and long sequences, HumanDiT [39] proposed a diffusion-based framework for pose-conditioned video generation. The model utilizes prefix latent reference strategies to effectively capture the subtle pose variations and details in long

video sequences, improving temporal consistency. Additionally, PGMM [40] developed a pose-guided motion model that generates fine-grained sign language videos through optical flow warping and pose adaptation modules, demonstrating advantages in generating precise and coherent actions.

BASELINE METHOD SELECTION

We select 5 publicly available representative methods for comparative evaluation, which reflect the latest development trends in video generation. Among them, VideoLDM adopts a latent diffusion model architecture specifically optimized for video generation tasks. VideoPD is based on a probabilistic diffusion framework, showing excellent performance in video quality and temporal consistency. VideoFusion enhances video generation control capability by fusing multiple modal information. SignDiff is a novel method specifically for sign language video generation, adopting ControlNet architecture to achieve controllable generation, demonstrating unique advantages in controlled generation contexts. SignGen, as the current state-of-the-art end-to-end sign language video generation method, uses latent diffusion models to directly generate sign language videos from text, representing the highest level in this field. The selection of these baseline methods fully reflects the technological development trajectory from general video generation to specialized sign language generation, providing comprehensive performance comparison benchmarks for our method.

DATASET AND PREPROCESSING.

We conduct comprehensive experimental evaluation on the Phoenix-2014.v3 dataset, which is one of the most authoritative benchmark datasets in German sign language recognition and generation. The dataset contains approximately 1000 sentences of high-quality sign language videos, mainly covering common sign language expressions in weather forecast scenarios, with each video annotated with detailed German text and corresponding sign language vocabulary sequences. For video data preprocessing, we adopt a standardized processing pipeline to ensure training consistency and stability. Original videos are first resized to the model-recommended resolution of 480×832, which maintains detailed information of sign language gestures while considering model generation quality. The model uses a fixed length of 81 frames as input sequence, projecting videos to latent space through a pre-trained VAE encoder, obtaining latent representations with dimensions [21, 16, 60, 104]. This design achieves an 8× downsampling ratio, significantly reducing computational complexity while preserving important spatial and temporal information.

ABLATION STUDY

a) *Effectiveness of Multi-Granular Control*: The results in Table II demonstrate the complementary roles of different control modalities in sign language video generation. Using a single control signal already yields reasonable performance, with pose clearly providing the strongest semantic guidance, as reflected by higher BLEU and ROUGE-L scores compared to

TABLE II

ABLATION ON THE EFFECTIVENESS OF MULTI-GRANULAR CONTROL. POSE PROVIDES THE STRONGEST SEMANTIC GUIDANCE, WHILE FLOW AND DEPTH FURTHER IMPROVE TEMPORAL COHERENCE AND STRUCTURAL CONSISTENCY.

Pose	Flow	Depth	BLEU↑	ROUGE-L↑	SSIM↑	FVD↓
✓	✗	✗	19.9	45.5	0.741	610
✗	✓	✗	18.7	43.6	0.736	655
✗	✗	✓	18.2	42.9	0.731	690
✓	✓	✗	20.8	46.4	0.752	565
✓	✗	✓	20.4	46.0	0.755	575
✗	✓	✓	19.4	44.7	0.748	605
✓	✓	✓	21.8	47.3	0.760	523

TABLE III

ABLATION ON THE EFFECTIVENESS OF THE CONTROL DECAY STRATEGY. SMOOTH SCHEDULES (COSINE/SIGMOID) TEND TO BE SLIGHTLY MORE STABLE THAN ABRUPT DECAY (STEP), IMPROVING ROBUSTNESS UNDER INCOMPLETE CONTROL AT INFERENCE.

Decay schedule	BLEU↑	ROUGE-L↑	SSIM↑	LPIPS↓	FVD↓
No decay (always full control)	21.1	46.6	0.754	0.029	548
Linear decay	21.8	47.3	0.760	0.025	523
Cosine decay	22.0	47.5	0.761	0.025	518
Step decay	21.6	47.0	0.757	0.027	532
Sigmoid decay	21.9	47.4	0.761	0.025	520

flow or depth alone. Optical flow and depth, while weaker in isolation, contribute important temporal and structural cues, respectively. Combining two controls consistently improves all metrics over single-control settings, indicating that semantic accuracy and visual coherence benefit from multi-source conditioning. The full combination of pose, flow, and depth achieves the best overall performance, reducing FVD substantially while simultaneously improving SSIM and language consistency metrics, validating the effectiveness of the proposed multi-granular control design.

b) Effectiveness of Control Decay Strategy: Table III analyzes the impact of different control decay schedules on robustness under incomplete control at inference. Without decay, the model tends to overfit to fully available controls, leading to inferior generalization. Introducing gradual decay consistently improves both semantic accuracy and visual quality, with smooth schedules such as cosine and sigmoid decay yielding the most balanced gains across metrics. These strategies outperform abrupt step decay, which slightly degrades temporal stability as reflected by higher FVD and LPIPS. Overall, the results indicate that smooth control decay better regularizes training, enabling the model to rely on partial control signals and text semantics more effectively, thereby enhancing robustness and practical usability.

FUTURE WORK

For future work, several directions are worth exploring. First, extending the framework to multi-signer and cross-lingual sign language generation could broaden its applicability, enabling adaptation to different signing styles, regional variations, and multiple sign languages while preserving

semantic consistency. Second, explicitly incorporating non-manual signals—such as facial expressions, mouth shapes, and gaze direction—with precise temporal alignment may enhance semantic fidelity, given their critical grammatical and emotional roles. Third, designing adaptive control selection mechanisms that dynamically adjust modality reliance based on input quality could improve robustness in real-world scenarios where certain control signals are noisy or missing. Finally, leveraging larger-scale datasets and exploring multi-task learning with related tasks such as sign language recognition or translation could further boost the model’s generalization and transferability, while continual learning strategies may allow long-term adaptation to evolving linguistic patterns without full retraining.